

Reanalysis of Volker (2022)

Network embeddedness, trust, and cooperation

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Overview

- Reproduces Table 2 of the paper.
- Source: Volker, T. B. (2022). *The future is made today: Concerns for reputation foster trust and cooperation* [Unpublished master's thesis]. Utrecht University. PDF
- 16 study-specific tests from 8 studies, each on a regression coefficient β :
 - $H_1 : \beta > 0$
 - $H_u : \beta \in \mathbb{R}$
 - $H_c : \beta \leq 0$
- Complexity $c_1^{(t)} = 0.5$ in all tests: the prior $N(0, \hat{\Sigma}_\beta/b)$ is centered at 0 (Volker, 2022, pp. 32, 36).

```
library(dplyr)
library(here)

# Function that enumerates all components of the complete joint complement
source(here("functions/compute_joint_complements.r"))
```

```

# Helpers to format numbers in the text
fmt2 <- function(x) sprintf("%.2f", x) # 2 decimals
fmt4 <- function(x) sprintf("%.4f", x) # 4 decimals
fmtE <- function(x) {
  if (!is.finite(x)) return("\\infty")
  e <- floor(log10(abs(x)))
  sprintf("%.2f \\times 10^{%d}", x / 10^e, e)
}

```

Data

- Layout from Volker's Table 2; Bayes factors from his Table 4 (3 decimals).
- Trustfulness: proportion of the endowment sent to the trustee.
- Trustworthiness: proportion of the amount sent that is returned (p. 27).

```

table2_df <- data.frame(
  study = c(
    rep("bolton_2004", 2),
    rep("duffy_2013_no_netmin", 2),
    rep("duffy_2013_no_netfull", 2),
    rep("buskens_2010", 2),
    rep("van_miltenburg_2012", 2),
    rep("frey_2019", 2),
    rep("barrera_buskens_2009", 2),
    "seinen_schram_2006",
    "corten_2016"
  ),
  outcome = c(
    rep(c("Trustfulness", "Trustworthiness"), 7),
    "Cooperation (helping)",
    "Cooperation"
  ),
  no_network = c(
    0.67, 0.69, 0.62, 0.68, 0.75, 0.62, 0.83, 0.87,
    0.90, 0.90, 0.77, 0.67, 0.73, 0.46, 0.38, 0.49
  ),
  network = c(
    0.75, 0.61, 0.64, 0.77, 0.86, 0.94, 0.94, 0.94,
    0.88, 0.91, 0.81, 0.88, 0.70, 0.50, 0.61, 0.37
  ),
  bf_iu = c(
    1.474, 0.642, 1.292, 1.823, 1.799, 2.000, 1.848,
    1.680, 0.675, 1.115, 1.426, 1.998, 0.482, 1.450,
    1.902, 0.102
  ),
  bf_ic = c(
    2.799, 0.473, 1.825, 10.319, 8.958, 2.60e+04,
    12.168, 5.255, 0.509, 1.260, 2.486, 1.01e+03,
    0.318, 2.634, 19.351, 0.054
  )
)

```

- This is a check: in the special case where $c_1^{(t)} = 0.5$ for all studies, the result should equal the one from equal weighting.
- Volker reports $BF_{1,u}^{(t)}$ and $BF_{1,c}^{(t)}$.
- `compute_joint_complements()` also needs $BF_{c,u}^{(t)}$, which can be obtained by transitivity: $BF_{c,u}^{(t)} = BF_{1,u}^{(t)} / BF_{1,c}^{(t)}$.

```
results_df <- table2_df %>%
  transmute(
    Study = study,
    Outcome = outcome,
    bf_1u = bf_iu,
    bf_1c = bf_ic,
    bf_cu = bf_iu / bf_ic
  )

# Input for compute_joint_complements():
# P_theta_in_Hc is the complexity of the complement, 1 - c_1 = 0.5
table2_ready <- results_df %>%
  select(Study, bf_1u, bf_cu) %>%
  mutate(P_theta_in_Hc = 0.5)

T <- nrow(results_df) # number of study-specific tests (16)
```

Study	Outcome	$BF_{1,u}$	$BF_{1,c}$	$BF_{c,u}$
bolton_2004	Trustfulness	1.474	2.799	0.527
bolton_2004	Trustworthiness	0.642	0.473	1.357
duffy_2013_no_netmiff	Trustfulness	1.292	1.825	0.708
duffy_2013_no_netmiff	Trustworthiness	1.823	10.319	0.177
duffy_2013_no_netfull	Trustfulness	1.799	8.958	0.201
duffy_2013_no_netfull	Trustworthiness	2.000	26000.000	0.000
buskens_2010	Trustfulness	1.848	12.168	0.152
buskens_2010	Trustworthiness	1.680	5.255	0.320
van_miltenburg_2012	Trustfulness	0.675	0.509	1.326
van_miltenburg_2012	Trustworthiness	1.115	1.260	0.885
frey_2019	Trustfulness	1.426	2.486	0.574
frey_2019	Trustworthiness	1.998	1010.000	0.002
barrera_buskens_2009	Trustfulness	0.482	0.318	1.516
barrera_buskens_2009	Trustworthiness	1.450	2.634	0.550
seinen_schram_2006	Cooperation (helping)	1.902	19.351	0.098
corten_2016	Cooperation	0.102	0.054	1.889

Product Bayes factors

```
bf_1u <- prod(results_df$bf_1u) # against the joint unconstrained hypothesis
bf_1c <- prod(results_df$bf_1c) # against the incomplete joint complement
```

- $BF_{1,u}^{(\text{joint})} = 7.24$ (Volker: 7.27; difference due to rounding).
- $BF_{1,c}^{(\text{joint})} = 5.23 \times 10^{11}$.

BF against the complete joint complement

- Complete joint complement: $H_1^{(t)}$ is false in at least one test.
- Computed in three ways; with exact inputs, all three agree here (with $c_1^{(t)} = 0.5$, equal and complexity weighting coincide).

1: Enumeration (Eq. 7)

- Sums over all $2^T - 1 = 65,535$ components with equal weights, giving $\text{BF}_{c^*,u}^{(\text{joint})}$.
- Then $\text{BF}_{1,c^*}^{(\text{joint})} = \text{BF}_{1,u}^{(\text{joint})} / \text{BF}_{c^*,u}^{(\text{joint})}$.

```
res <- compute_joint_complements(table2_ready,
                                replace_0_with = 0.001,
                                round_res = FALSE)

# Sanity check: BF_1u(joint) / BF_cu(joint) should equal the product of the BF_1c's
all.equal(res["BES_c", "bf_1u"] / res["BES_c", "bf_cu"], bf_1c)
```

```
[1] TRUE
```

```
bf_1c_star_enumeration <- res["BES_c", "bf_1u"] / res["BES_c_star", "bf_cu_star"]
```

2: Complexity weighting (Eq. 8)

- Check: in the special case where $c_1^{(t)} = 0.5$ for all studies, the result should equal the one obtained from equal weighting.
- Uses the $\text{BF}_{1,u}^{(t)}$ and complexities $c_1^{(t)}$.

$$\text{BF}_{1,c^*}^{(\text{joint})} = \frac{\prod_{t=1}^T \text{BF}_{1,u}^{(t)} (1 - \prod_{t=1}^T c_1^{(t)})}{1 - \prod_{t=1}^T c_1^{(t)} \text{BF}_{1,u}^{(t)}}$$

```
c1 <- 1 - table2_ready$P_theta_in_Hc # complexity c_1^(t) of H_1 in each test

bf_1c_star_formula_complexity <-
  prod(results_df$bf_1u) * (1 - prod(c1)) / (1 - prod(c1 * results_df$bf_1u))
```

3: Equal weighting (Eq. 9)

- Uses only the $\text{BF}_{1,c}^{(t)}$.

$$\text{BF}_{1,c^*}^{(\text{joint})} = \frac{2^T - 1}{\prod_{t=1}^T (1 + 1/\text{BF}_{1,c}^{(t)}) - 1}$$

```
bf_1c_star_formula_equal <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)
```

Comparison

Method	Computed from	$\text{BF}_{1,c^*}^{(\text{joint})}$
Enumeration (Eq. 7)	$\text{BF}_{1,u}^{(t)}$	7.2397
Complexity weighting (Eq. 8)	$\text{BF}_{1,u}^{(t)}$ and $c_1^{(t)}$	7.2396
Equal weighting (Eq. 9)	$\text{BF}_{1,c}^{(t)}$	7.2795

- Enumeration and Eq. 8: 7.24; Eq. 9: 7.28.
- The difference comes from *which* rounded Bayes factors each method uses:
 - Enumeration: numerator $\prod_t \text{BF}_{1,u}^{(t)}$; denominator $\text{BF}_{c^*,u}^{(\text{joint})} \approx 1$.
 - Eq. 8: $\text{BF}_{1,u}^{(t)}$.
 - Eq. 9: $\text{BF}_{1,c}^{(t)}$.

Rounding check

- **Goal:** How much of the gap between $\text{BF}_{1,u}^{(\text{joint})} = 7.24$ and $\text{BF}_{1,c^*}^{(\text{joint})} = 7.28$ (Eq. 9) is due to rounding?
- The two are computed from different columns ($\text{BF}_{1,u}$ vs. $\text{BF}_{1,c}$), which Volker rounded separately to 3 decimals.
- **Logic:** With $c_1 = 0.5$, each $\text{BF}_{1,u}^{(t)}$ determines its $\text{BF}_{1,c}^{(t)}$ exactly, and vice versa:

$$\text{BF}_{1,c}^{(t)} = \frac{\text{BF}_{1,u}^{(t)}}{2 - \text{BF}_{1,u}^{(t)}} \iff \text{BF}_{1,u}^{(t)} = \frac{2 \text{BF}_{1,c}^{(t)}}{1 + \text{BF}_{1,c}^{(t)}}$$

- Set 1: reported $\text{BF}_{1,u}^{(t)} \rightarrow$ implied $\text{BF}_{1,c}^{(t)}$.
- Set 2: reported $\text{BF}_{1,c}^{(t)} \rightarrow$ implied $\text{BF}_{1,u}^{(t)}$.
- Within a set, both joint BF's use the same information, so any difference is genuine (not due to rounding).
- **Expectation**
- With $c_1^{(t)} = 0.5$, Eqs. 8 and 9 reduce to $\text{BF}_{1,c^*}^{(\text{joint})} = \text{BF}_{1,u}^{(\text{joint})} \times \text{factor}$, where $\text{factor} = (2^T - 1)/(2^T - \text{BF}_{1,u}^{(\text{joint})})$.

```
# Step 1: ratio BF_1c*(joint) / BF_1u(joint) implied by Eqs. 8 and 9 when c_1 = 0.5
factor <- (2^T - 1) / (2^T - bf_1u)

# Step 2: expected BF_1c*(joint) without rounding
bf_1cstar_exp <- bf_1u * factor

# Step 3: expected difference between BF_1c*(joint) and BF_1u(joint)
diff_exp <- bf_1cstar_exp - bf_1u
```

With $T = 16$ and $\text{BF}_{1,u}^{(\text{joint})} = 7.238915$:

1. **Factor:** how much larger $\text{BF}_{1,c^*}^{(\text{joint})}$ should be than $\text{BF}_{1,u}^{(\text{joint})}$ (here 0.0095% larger):

$$\text{factor} = \frac{65535}{65536 - 7.238915} = 1.000095209$$

2. **Expected** $\text{BF}_{1,c^*}^{(\text{joint})}$ without rounding:

$$7.238915 \times 1.000095209 = 7.239604$$

3. **Expected within-set difference:**

$$7.239604 - 7.238915 = 0.000689 \approx 0.0007$$

- This is the genuine difference that remains without rounding. The table below compares it with the observed differences.

```
# Set 1: consistent set based on the reported BF_1u's
bf_1c_from_1u <- results_df$bf_1u / (2 - results_df$bf_1u)

check_set1_bf_1u_joint <- prod(results_df$bf_1u)
check_set1_bf_1cstar <- (2^T - 1) / (prod(1 + 1 / bf_1c_from_1u) - 1)

# Set 2: consistent set based on the reported BF_1c's
bf_1u_from_1c <- 2 * results_df$bf_1c / (1 + results_df$bf_1c)

check_set2_bf_1u_joint <- prod(bf_1u_from_1c)
check_set2_bf_1cstar <- (2^T - 1) / (prod(1 + 1 / results_df$bf_1c) - 1)
```

Inputs	$\text{BF}_{1,u}^{(\text{joint})}$	$\text{BF}_{1,c^*}^{(\text{joint})}$	Difference
Set 1: reported $\text{BF}_{1,u}$ values	7.2389	7.2396	0.0007
Set 2: reported $\text{BF}_{1,c}$ values	7.2788	7.2795	0.0007
Reported values (mixed sets)	7.2389	7.2795	0.0406

Conclusion:

- Within each set: difference ≈ 0.0007 , as expected.
- Reported (mixed) values: difference = 0.0406.
- So the gap is almost entirely due to rounding, mostly from the smallest Bayes factors (e.g., Corten et al., 2016: reported $\text{BF}_{1,u} = 0.102$, but $\text{BF}_{1,c} = 0.054$ implies $\text{BF}_{1,u} = 0.1025$).

Summary

$\text{BF}_{1,u}^{(\text{joint})}$	7.24
$\text{BF}_{1,c}^{(\text{joint})}$	5.23e+11
$\text{BF}_{1,c^*}^{(\text{joint})}$	7.28

Session info

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
```

```
Platform: aarch64-apple-darwin23
```

```
Running under: macOS Tahoe 26.5.2
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK version 3.
```

```
locale:
```

```
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
```

```
time zone: Europe/Berlin
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices datasets  utils      methods    base
```

```
other attached packages:
```

```
[1] stringr_1.6.0 tibble_3.3.1 here_1.0.2    dplyr_1.2.1
```

```
loaded via a namespace (and not attached):
```

```
[1] vctrs_0.7.3      cli_3.6.6        knitr_1.52       rlang_1.3.0  
[5] xfun_0.61        stringi_1.8.9    ote1_0.2.0       renv_1.2.4  
[9] generics_0.1.4   jsonlite_2.0.0   glue_1.8.1       rprojroot_2.1.1  
[13] htmltools_0.5.9  sass_0.4.10      rmarkdown_2.32   evaluate_1.0.5  
[17] jquerylib_0.1.4  fastmap_1.2.0    yaml_2.3.12      lifecycle_1.0.5  
[21] compiler_4.6.1   pkgconfig_2.0.3  digest_0.6.39    R6_2.6.1  
[25] tidyselect_1.2.1 pillar_1.11.1    magrittr_2.0.5   bslib_0.12.0  
[29] withr_3.0.3      tools_4.6.1      cachem_1.1.0
```